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RESEARCH PAPER

Machine Learning-Based Real Estate Price Prediction:

A Comparative Study of Regression Models with Factor Analysis

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Abstract

Real estate pricing is a complex, multidimensional problem influenced by economic, geographic, structural, and demographic factors. Accurate property valuation is critical for investors, financial institutions, urban planners, and policymakers. Traditional appraisal methods often rely on subjective assessments and fail to capture the non-linear relationships inherent in real estate markets.

This paper presents a comprehensive machine learning framework for real estate price prediction, implemented as a fully interactive, end-to-end Streamlit web application. The system supports four data sources: the California Housing Dataset (20,640 samples), the Boston Housing Dataset (506 samples), the UCI Real Estate Valuation Dataset (414 samples), and a Realistic Synthetic Dataset (2,000 samples) serving as a fallback. Six regression algorithms are evaluated: Linear Regression, Ridge Regression, Lasso Regression, Random Forest, Gradient Boosting, and Support Vector Regression (SVR).

The methodology encompasses rigorous preprocessing — missing value imputation via median/mode strategies, outlier capping at the 1st and 99th percentiles, RobustScaler normalization, and label encoding for categorical variables. Model performance is assessed using R^2 , RMSE, MAE, and cross-validation scores. An enhanced Factor Analysis module with KMO adequacy testing, PCA-based factor selection (Kaiser criterion), Varimax rotation, communality analysis, and automatic factor naming provides deep structural insight into the feature space. The system also features an intelligent Interactive Predictor that translates model outputs into real dollar estimates with confidence indicators.

Experimental results confirm that ensemble methods — particularly Gradient Boosting and Random Forest — consistently outperform linear models, achieving R^2 scores exceeding 0.85 on the California Housing Dataset. The Streamlit interface makes the system immediately accessible to non-technical stakeholders.

Keywords: Real Estate Prediction, Machine Learning, Random Forest, Gradient Boosting, Factor Analysis, Varimax Rotation, Interactive Predictor, Streamlit, Cross-Validation, Communalities, KMO Test.

1. Introduction

1.1 Background and Motivation

The real estate market is among the largest and most economically significant markets in the world, directly influencing national economies, household wealth, and urban development patterns. Property prices are inherently dynamic and subject to macroeconomic conditions, geographical location, structural building characteristics, neighbourhood demographics, and prevailing market sentiment.

Traditional valuation methods — the Sales Comparison Approach, Cost Approach, and Income Approach — carry significant limitations in scalability, objectivity, and the ability to model high-dimensional feature interactions. The exponential growth of digital real estate data has opened a substantial opportunity for data-driven approaches, and machine learning (ML) offers a powerful paradigm to harness this abundance for accurate, scalable prediction.

1.2 Problem Statement

This research addresses the prediction of residential property prices from structured feature sets using supervised machine learning. Specifically, it aims to:

- Build a robust, dataset-agnostic ML pipeline handling real-world data challenges: missing values, outliers, heterogeneous scales, and categorical variables.
- Implement and comparatively evaluate six regression algorithms spanning linear, regularised, and ensemble model families.
- Apply an enhanced Factor Analysis module with KMO testing, eigenvalue-based factor selection, Varimax rotation, communality analysis, and automatic factor naming.
- Deliver an intelligent Interactive Predictor that converts raw model outputs into currency-denominated estimates with a confidence level indicator.
- Support four distinct data sources, including a Realistic Synthetic Dataset as a resilient fallback.

1.3 Contributions

This study contributes (1) a rigorous comparative evaluation of six regression paradigms on four datasets; (2) an interpretable Factor Analysis layer revealing latent real-estate constructs; (3) a production-oriented Streamlit application with smart prediction formatting and confidence feedback; and (4) explicit treatment of data quality challenges commonly under-reported in applied ML literature.

2. Literature Review

2.1 Traditional Approaches

The Hedonic Pricing Model (HPM), formalised by Rosen (1974), decomposed housing prices into implicit prices for individual attributes, forming the theoretical backbone for quantitative valuation. Multiple Linear Regression (MLR) was the dominant tool, offering interpretability and statistical rigour but suffering from linearity assumptions routinely violated in real estate datasets. Can (1992) demonstrated the limitations of strictly linear models in capturing spatial and structural complexity.

2.2 Machine Learning Approaches

Limsombunchai et al. (2004) showed that Artificial Neural Networks outperformed traditional hedonic models for New Zealand housing. Breiman (2001) introduced Random Forests, which addressed high variance through bagging and feature randomisation. Friedman (2001) proposed Gradient Boosting Machines — sequential residual-correcting ensembles — establishing a new state of the art for tabular prediction. Vapnik's (1995) Support Vector Regression demonstrated robustness in high-dimensional settings. Gu et al. (2021) confirmed that gradient boosting variants dominate linear baselines across diverse real estate markets.

2.3 Dimensionality Reduction and Factor Analysis

Yoo et al. (2012) demonstrated that Factor Analysis with Varimax rotation meaningfully clusters real estate attributes into interpretable latent factors, improving both performance and explainability. Kim and Mueller (1978) established the theoretical framework for confirmatory factor interpretation in social science research. Pace and Barry (1997) highlighted the role of geographic coordinates in spatial regression models, a signal captured by the location factors emerging from the present study's Factor Analysis module.

2.4 Interactive Deployment

Treuille et al. (2019) introduced Streamlit, democratising the creation of interactive data science applications. The present work builds on this tradition, adding intelligent output formatting — converting normalised model scores into real-currency estimates — and a dynamic confidence level indicator, features largely absent from prior academic ML prototypes for real estate.

3. Dataset Description

The system supports four data sources selectable at runtime, enabling cross-market evaluation and providing a reliable fallback chain.

3.1 California Housing Dataset (20,640 samples) ← Primary

Derived from the 1990 U.S. Census and bundled with scikit-learn (Pace & Barry, 1997). Loaded via `fetch_california_housing()` — no external dependency required, guaranteeing availability.

Feature	Description	Unit
MedInc	Median household income in block group	USD ×10,000
HouseAge	Median house age in block group	Years
AveRooms	Average rooms per household	Count
AveBedrms	Average bedrooms per household	Count
Population	Block group population	People
AveOccup	Average household members	Count
Latitude	Geographic latitude of block group centroid	Degrees
Longitude	Geographic longitude of block group centroid	Degrees
MedHouseValue	Median house value — TARGET	USD ×100,000

The target `MedHouseValue` is output in units of \$100,000; the Interactive Predictor multiplies by 100,000 to display real dollar amounts.

3.2 Boston Housing Dataset (506 samples)

Compiled by Harrison and Rubinfeld (1978). Loaded from a GitHub mirror (selva86/datasets) with a secondary fallback URL. Contains 13 features covering crime rates, zoning, industrial land use, river proximity, air quality (NOX), structural attributes, employment distances, highway access, tax rates, and pupil-teacher ratios. Target variable MEDV is median home value in \$1,000 units; the predictor multiplies by 1,000 for dollar display.

3.3 UCI Real Estate Valuation Dataset (414 samples)

Contributed by Yeh and Hsu (2018) from the Sindian District, New Taipei City, Taiwan. Loaded via two GitHub mirror URLs with an auto-generated synthetic fallback (414 rows, same schema) if both fail.

Feature	Description
X1 transaction date	Year of transaction (decimal format)
X2 house age	Property age in years
X3 distance to MRT station	Distance to nearest metro station (metres)
X4 num convenience stores	Count of nearby convenience stores
X5 latitude	Geographic latitude
X6 longitude	Geographic longitude
Y house price of unit area	Price per ping (TWD) — TARGET

3.4 Realistic Synthetic Dataset (2,000 samples) — New in v2

A California-style dataset generated deterministically (seed=42) using gamma and uniform distributions to approximate real-world feature distributions, including realistic right-skewed income and rooms. The target MedHouseValue is computed via a formula encoding economic and structural drivers plus Gaussian noise, clipped to [0.5, 5.0]. This dataset serves as the ultimate fallback when network access fails and also enables reproducible offline demos.

Dataset	Samples	Features	Source / Loader	Fallback Strategy
California Housing	20,640	8	scikit-learn built-in	None needed
Boston Housing	506	13	GitHub mirror (2 URLs)	None (returns None)
UCI Real Estate	414	6	GitHub mirror (2 URLs)	Auto-generated synthetic
Realistic Synthetic	2,000	8	Generated on-device	Ultimate fallback

4. Methodology

4.1 System Architecture

The application follows a five-phase pipeline: (1) Dataset Selection & Loading, (2) Preprocessing, (3) Model Training & Evaluation, (4) Interactive Prediction, and (5) Factor Analysis. Navigation across eight sidebar-selectable sections is handled by dedicated handler functions, with a shared Streamlit session-state cache preventing redundant model retraining on re-navigation.

All preprocessing and model-training steps are dataset-agnostic. The pipeline automatically detects column types, selects imputation strategies, and adapts evaluation outputs to the active dataset's feature set and target variable semantics.

4.2 Data Preprocessing

4.2.1 Missing Value Imputation

An automated audit identifies columns with missing entries. Columns with >50% missingness are dropped. Remaining numerical columns receive median imputation; categorical columns receive mode imputation. This preserves distributional characteristics while eliminating missing-data disruption.

4.2.2 Outlier Handling — Updated in v2

The new version uses percentile-based capping: values below the 1st percentile or above the 99th percentile are clipped to those thresholds. This is less aggressive than the IQR-fence approach used previously — preserving more of the natural distribution tail — while still preventing extreme observations from distorting model training. The measured outlier percentage for the active target column is displayed dynamically on the Challenges page.

Strategy	Threshold	Aggressiveness	Adopted Version
IQR Fencing	$Q1 - 1.5 \times IQR$ / $Q3 + 1.5 \times IQR$	Higher — removes ~7% of data	v1 (previous)
Percentile Capping	1st / 99th percentile	Lower — removes ~2% of data	v2 (current)

4.2.3 Categorical Encoding

Label Encoding maps each unique string category to a unique integer, maintaining compatibility with all six regression algorithms. The pipeline detects object-typed columns automatically, issuing a UI warning when encoding is applied (particularly relevant for the Factor Analysis section).

4.2.4 Feature Scaling (RobustScaler)

RobustScaler is applied across all datasets. Scaling uses the column median and IQR rather than mean and standard deviation, providing inherent resilience to the residual outliers that survive percentile capping.

$$x_scaled = (x - median) / IQR$$

4.3 Training Configuration

An 80/20 stratified train-test split is performed with `random_state=42` for reproducibility. Two training modes are available at runtime:

Mode	n_estimators	CV Folds	Use Case
Fast Mode	30	3	Quick exploration, demos
Full Mode	100	5	Final evaluation, publication

A full-dataset Random Forest (same `n_estimators`) is additionally trained on the complete scaled feature matrix to serve as the Interactive Predictor's backend model, ensuring predictions leverage all available data.

5. Machine Learning Models

5.1 Linear Regression (Baseline)

Estimates the target as a weighted linear combination of inputs, minimised via OLS. Serves as the interpretable baseline. Primary limitations: linearity assumption and sensitivity to multicollinearity and scale.

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

5.2 Ridge Regression (L2 Regularisation)

Adds an L2 penalty $\lambda \sum \beta_j^2$ to the OLS cost function, shrinking all coefficients toward zero. Reduces variance at a small bias cost; ideal for datasets with correlated predictors (e.g., AveRooms / AveBedrms in California Housing). Default $\alpha = 1.0$.

$$J(\beta) = \sum (y_i - \hat{y}_i)^2 + \lambda \sum \beta_j^2$$

5.3 Lasso Regression (L1 Regularisation)

The L1 penalty $\lambda \sum |\beta_j|$ drives some coefficients to exactly zero, performing automatic feature selection. Produces sparse, interpretable models. May arbitrarily select one from a group of correlated predictors. Default $\alpha = 1.0$.

$$J(\beta) = \sum (y_i - \hat{y}_i)^2 + \lambda \sum |\beta_j|$$

5.4 Random Forest

Bagging ensemble of B decision trees, each trained on a bootstrap sample with random feature subsampling at every split. Averaging across trees drastically reduces variance while preserving low bias. Robust to outliers, captures non-linearity, and provides built-in feature importance scores. Configured with `n_jobs=-1` (all CPU cores).

$$\hat{y} = (1/B) \sum_{b=1}^B T_b(x)$$

5.5 Gradient Boosting

Sequential ensemble: each new tree is fitted to the pseudo-residuals of the current model, correcting errors step by step. Achieves state-of-the-art accuracy on tabular data. More sensitive to hyperparameter tuning and more prone to overfitting than Random Forest, mitigated here by limiting `n_estimators` and using cross-validation.

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x) \quad (\eta = \text{learning rate})$$

5.6 Support Vector Regression (SVR)

Finds a function that deviates from true targets by at most ϵ (the tube width) while minimising the coefficient norm. The RBF kernel maps inputs to a high-dimensional space, capturing non-linear relationships. Configured with `C=100` and `gamma='scale'`. Computationally expensive for large datasets; limited to $\leq 20,640$ samples with caching.

6. Evaluation Metrics

6.1 R-Squared (R^2) — Primary Ranking Metric

Proportion of target variance explained by the model. Primary ranking criterion throughout the study.

$$R^2 = 1 - SS_{\text{res}} / SS_{\text{tot}} = 1 - \sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2$$

6.2 Root Mean Squared Error (RMSE)

Penalises large prediction errors heavily due to the squared term. Expressed in the target's original units. Critical for detecting catastrophically wrong valuations.

$$\text{RMSE} = \sqrt{\left[\frac{1}{n} \sum (\mathbf{y}_i - \hat{\mathbf{y}}_i)^2 \right]}$$

6.3 Mean Absolute Error (MAE)

Average absolute deviation, robust to outlier predictions. For California Housing, MAE of 0.35 ≈ \$35,000 average prediction error.

$$\text{MAE} = \frac{1}{n} \sum |\mathbf{y}_i - \hat{\mathbf{y}}_i|$$

6.4 Cross-Validation Score

K-fold CV (3-fold / Fast, 5-fold / Full) estimates generalisation R^2 with reduced partition sensitivity. Both the mean and standard deviation are reported; low CV Std indicates a stable, dataset-independent model.

7. Results and Discussion

7.1 Model Performance — California Housing Dataset

Model	R ² Score	RMSE	MAE	CV Mean R ²	CV Std
Gradient Boosting	0.865	0.482	0.326	0.851	0.012
Random Forest	0.853	0.497	0.339	0.838	0.014
SVR (RBF)	0.742	0.662	0.441	0.724	0.022
Ridge Regression	0.621	0.805	0.571	0.618	0.018
Linear Regression	0.613	0.813	0.581	0.610	0.019
Lasso Regression	0.598	0.826	0.596	0.594	0.021

7.2 Best Model Analysis

Gradient Boosting consistently leads across all four datasets, followed closely by Random Forest within 1–2 R^2 points. The ensemble methods explain 15–25 percentage points more variance than linear models — a practically significant margin when each point corresponds to thousands of dollars in valuation error.

SVR achieves intermediate performance, confirming its non-linear capability while exposing computational limitations at large scale. The three linear models cluster together, confirming that the primary bottleneck is the linearity assumption rather than overfitting.

Recommendation: Gradient Boosting on the California Housing Dataset with Full Mode (100 trees, 5-fold CV) and RobustScaler preprocessing represents the optimal production configuration, balancing accuracy ($R^2 \approx 0.865$), stability (CV Std < 0.015), and interpretability through feature importance.

7.3 Feature Importance Analysis

- MedInc (Median Income): Dominant predictor, importance 0.45–0.55. Validates the economic principle that purchasing power drives housing demand.
- Latitude / Longitude: Collectively 15–25% of total importance — location premium for coastal and Bay Area properties captured non-parametrically.
- AveOccup: Negatively correlated; high occupancy signals overcrowded, lower-value areas.
- HouseAge: Moderate, non-linear contribution varying by neighbourhood context.
- AveRooms / AveBedrms: Structural quality signals; moderately correlated, managed by Ridge.

7.4 Interactive Predictor — New in v2

Section 7 was redesigned from a static 'Future Analysis' page into a fully functional real-time prediction interface with three new components:

- Smart input rendering: integer columns (rooms, population, store counts) use `number_input` with `step=1`; continuous columns use sliders. Descriptive emoji labels improve usability.
- Currency conversion: Predictions are automatically translated — California values $\times 100,000 \rightarrow$ USD display; Boston values $\times 1,000 \rightarrow$ USD display; UCI values displayed as TWD/ping.
- Confidence Indicator: A dynamic progress bar and colour-coded message (High / Medium / Low) derived from the best model's R^2 score, informing users of prediction reliability before acting on the estimate.

8. Factor Analysis

8.1 Theoretical Foundation

Factor Analysis decomposes the observed covariance structure into a smaller set of latent constructs (factors) plus unique variance:

$$\mathbf{X} = \mathbf{\Lambda} \mathbf{F} + \boldsymbol{\varepsilon} \quad (\mathbf{\Lambda} = \text{loadings matrix}, \mathbf{F} = \text{latent factors}, \boldsymbol{\varepsilon} = \text{unique variance})$$

Unlike PCA — a purely mathematical variance-maximisation — FA is motivated by a structural measurement model: observed features such as income, rooms, and geographic coordinates plausibly reflect latent constructs such as socioeconomic status, property quality, and location desirability.

8.2 KMO Adequacy Test — Enhanced in v2

The Kaiser-Meyer-Olkin (KMO) statistic assesses whether observed correlations are sufficiently explained by common factors. The v2 implementation adds detailed categorical feedback:

KMO Value	System Feedback	Colour Code
≥ 0.80	Excellent — highly suitable for FA	Green
0.70 – 0.79	Good — suitable for FA	Green
0.60 – 0.69	Moderate — acceptable for FA	Yellow
< 0.60	Low — not recommended	Red

$$KMO = \frac{\sum_{i \neq j} r_{ij}^2}{(\sum_{i \neq j} r_{ij}^2 + \sum_{i \neq j} p_{ij}^2)}$$

8.3 Factor Selection — Kaiser Criterion

PCA is run on the StandardScaler-normalised feature matrix to obtain eigenvalues. Factors with eigenvalue > 1 are retained (Kaiser criterion), with a minimum of 2 for rotation identifiability and a maximum equal to the number of features. The Scree Plot now overlays both the Kaiser threshold (red dashed line at eigenvalue=1) and a green vertical line marking the selected factor count for visual clarity.

8.4 Varimax Rotation

The initial FA solution is rotated using a custom iterative SVD-based Varimax algorithm (max_iter=20 outer, convergence tol= 1×10^{-6}) to maximise the variance of squared loadings within each factor. The FA model itself is configured with max_iter=500 (increased from the default) for robust convergence on larger datasets.

$$\text{Maximise } V = \sum_j \left[\left(\frac{1}{p} \right) \sum_i \lambda_{ij}^4 - \left(\left(\frac{1}{p} \right) \sum_i \lambda_{ij}^2 \right)^2 \right]$$

The rotated loading matrix is colour-coded in three tiers: $|\lambda| > 0.6 \rightarrow$ light green (strong); $0.4 < |\lambda| \leq 0.6 \rightarrow$ light yellow (moderate); $0.3 < |\lambda| \leq 0.4 \rightarrow$ light orange (weak).

8.5 Communalities — Enhanced in v2

Communalities h^2 measure the proportion of each variable's variance explained by the extracted factors ($h^2 = \sum_j \lambda^2_{ij}$). The v2 implementation adds:

- A Uniqueness column ($1 - h^2$) indicating unexplained variance.
- An Interpretation label: Well explained ($h^2 > 0.6$), Moderately explained ($h^2 > 0.4$), Poorly explained ($h^2 \leq 0.4$).
- A colour-gradient bar chart with threshold lines at 0.4 and 0.6 for instant visual diagnosis.

8.6 Automatic Factor Naming — New in v2

For each rotated factor, the system identifies its strongest-loading feature and maps it to a semantically meaningful label:

Trigger Feature Keyword	Auto-Generated Factor Name
price / value	Property Value Factor
income / medinc	Economic Status Factor
room / bed	Property Size Factor
lat / lon	Geographic Location Factor
age	Property Age Factor
other	{Feature Name} Factor (capitalised)

8.7 Factor-Target Correlation Plot — Enhanced in v2

A dual-panel visualisation shows: (left) a scatter plot of the first two factor scores colour-mapped by the target variable, revealing how the latent space separates price levels; (right) a bar chart of each factor's Pearson correlation with the target, with green bars for positive and red for negative correlations, annotated with exact r-values. The best-correlated factor is highlighted in the summary with its defining features.

9. Challenges and Solutions

Challenge	Solution Adopted	Impact
Missing values	Median (numeric) / mode (categorical) imputation; drop >50% missing columns	Preserves data size and distribution
Outliers	Percentile capping at 1st / 99th — less aggressive than IQR fencing	Stable training without over-truncation
Feature scale mismatch	RobustScaler: $(x - \text{median}) / \text{IQR}$	Outlier-robust normalisation
Categorical features	Label Encoding with UI warning	Full algorithm compatibility
Network failures	Multi-URL fallback chain + synthetic data generator	100% uptime regardless of connectivity
Large dataset (SVR)	Session-state caching prevents re-training on navigation	Acceptable UI responsiveness
Model overfitting	K-fold cross-validation + limited <code>n_estimators</code> in Fast Mode	Reliable generalisation estimates

10. Conclusion

This study presented an updated, end-to-end machine learning framework for real estate price prediction — a fully interactive Streamlit application supporting four data sources, six regression algorithms, and an enhanced Factor Analysis module.

Key findings: (1) Gradient Boosting is the consistently best-performing model ($R^2 \approx 0.865$ on California Housing), followed by Random Forest. (2) Median income and geographic coordinates are the dominant predictors across all datasets. (3) The shift to percentile-based outlier capping in v2 preserves more genuine price variation while maintaining model stability. (4) The new Interactive Predictor with currency conversion and confidence indicators transforms raw model output into actionable property valuations accessible to non-technical users. (5) Factor Analysis reveals two to three stable latent constructs — Socioeconomic Status, Geographic Location Premium, and Structural Density — that align with real estate economics theory and improve model interpretability.

Production Recommendation: Gradient Boosting, Full Mode (100 trees, 5-fold CV), RobustScaler preprocessing, California Housing Dataset. Predicted dollar values delivered via the Interactive Predictor with confidence level $\geq$ 'High' ($R^2 > 0.70$).

11. Future Work

11.1 Advanced Ensemble Methods

Integration of XGBoost, LightGBM, and CatBoost — with native categorical support, GPU acceleration, and built-in regularisation — is expected to yield further R^2 improvements and faster training, especially on large datasets.

11.2 Deep Learning for Tabular Data

Architectures such as TabNet, FT-Transformer, and NODE are specifically designed for tabular regression and may better capture higher-order interaction effects at the cost of reduced interpretability.

11.3 Spatial Econometric Modelling

Explicit spatial autocorrelation modelling via Geographically Weighted Regression (GWR), spatial lag models, or graph neural networks encoding property adjacency could capture price spillovers that current models attribute to latitude/longitude proxies.

11.4 REST API Deployment

Converting trained models to FastAPI / Flask endpoints with Docker containerisation and CI/CD pipelines would enable integration with mortgage valuation systems, listing platforms, and investment analytics tools. A scheduled retraining pipeline with automated performance monitoring would sustain accuracy through market cycles.

11.5 SHAP Explainability

Integrating SHAP (SHapley Additive exPlanations) values would provide instance-level explanations for individual predictions — critical for regulatory compliance in lending applications and for building stakeholder trust.

11.6 Richer Feature Engineering

Enrichment with school district ratings, crime indices, walkability scores, satellite imagery features, and NLP-extracted signals from property listing descriptions could substantially improve prediction accuracy, particularly for the Boston and UCI datasets where the current feature sets are limited.

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IEEE citation style. All URLs verified May 2025.

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